FIXED INCOME PRIMER

Debt & Bonds

Features of Debt • Meaning & Characteristics of Bonds • Indian & US Government Bond Examples • Interest Rate Risk

A concept-to-application walkthrough with worked examples and charts

Features of Debt

Core traits that define any borrowing arrangement

Obligation to Repay

Borrower is legally bound to return the principal amount borrowed.

Interest Payment

Lender is compensated with interest — fixed or floating rate.

Fixed Tenure

A defined maturity date after which the loan is fully repaid.

Priority Over Equity

Debt holders are paid before shareholders, even in liquidation.

No Ownership Rights

Lenders gain no equity stake or voting rights in the borrower.

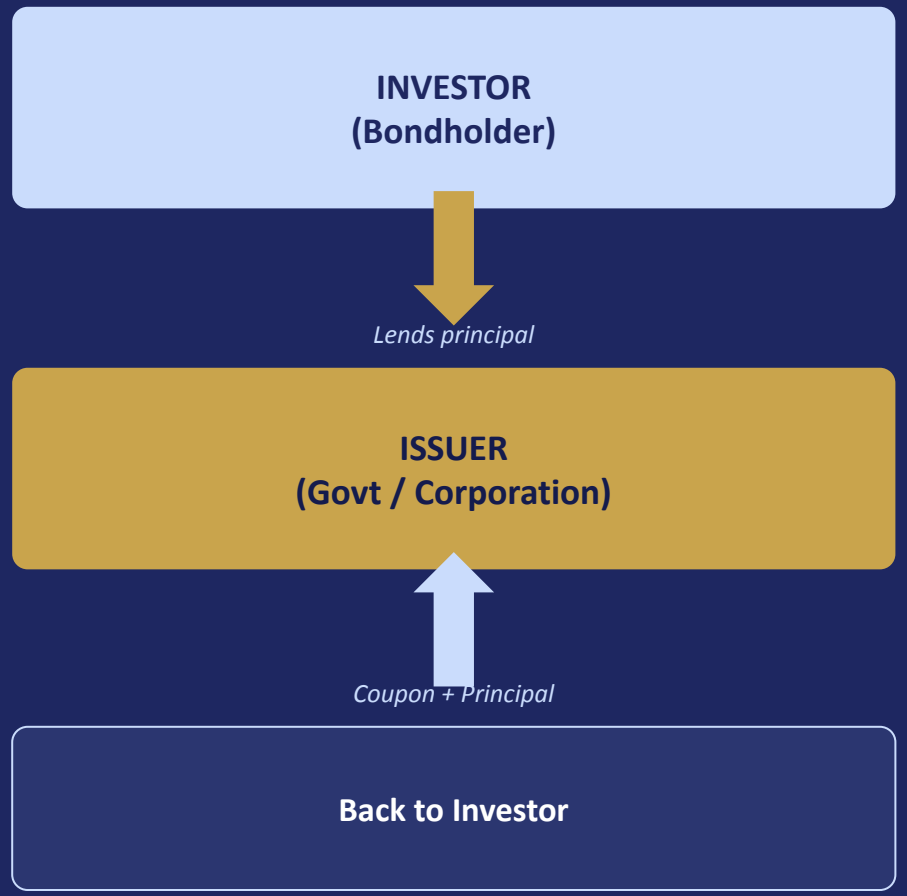
Collateral / Security

Debt may be secured by assets or issued on an unsecured basis.

Meaning of a Bond

A bond is a fixed-income security representing a loan made by an investor (the bondholder) to a borrower — typically a government, municipality, or corporation.

In simple terms: buying a bond means lending money to the issuer in exchange for periodic interest and the return of principal at a set future date.



Characteristics of a Bond

The defining terms every bond carries



Face / Par Value

Amount repaid at maturity; base for coupon calculation.



Coupon Rate

Fixed interest rate paid periodically, as % of face value.



Maturity Date

The date the principal is fully repaid and the bond ends.



Market Price

Trades at par, premium, or discount depending on yields.



Yield to Maturity

Effective annualized return if held until maturity.



Credit Rating

Agency assessment of issuer's ability to repay (risk grade).



Bond Type

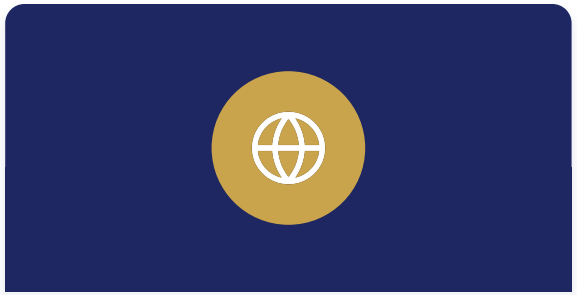
Secured/unsecured, fixed/floating, convertible, callable, zero-coupon.



Indenture

Legal contract defining issuer and bondholder obligations.

Types of Bonds — By Issuer



Government (Sovereign)

Issued by central governments (e.g., G-Secs, US Treasuries). Virtually zero default risk — lowest yields.



Municipal

Issued by local governments for infrastructure; may carry tax benefits in some countries.



Corporate

Issued by companies to raise capital; higher risk and yield, rated by agencies.



PSU / Public Sector

Issued by government-owned enterprises; moderate risk, often implicitly backed.

Types of Bonds — By Structure & Features

By Security

Secured
Backed by collateral / specific assets

Unsecured (Debentures)
No collateral; relies on issuer creditworthiness

By Rate Structure

Fixed-Rate
Coupon stays constant through the term

Floating-Rate
Coupon resets with a benchmark rate

Zero-Coupon
No periodic interest; issued at deep discount

Special Features

Convertible
Convertible into equity shares

Callable / Puttable
Issuer/holder can redeem early

Inflation-Indexed
Principal adjusts with inflation (CPI)

Perpetual
No maturity date; pays interest indefinitely

Bond Pricing: Present Value of Cash Flows

A bond's price is the present value of every future coupon plus the face value, discounted at the required yield.

$$P = \sum C / (1+r)^t + F / (1+r)^n$$

- P — Price of the bond today
- C — Periodic coupon payment (Face × Coupon Rate)
- r — Required rate of return / discount rate per period
- n — Number of periods until maturity
- F — Face value, redeemed at maturity

Par, Premium, or Discount?

The relationship between coupon rate and market yield sets the price



AT PAR

Coupon Rate = Market Yield

Price = Face Value



AT PREMIUM

Coupon Rate > Market Yield

Price > Face Value



AT DISCOUNT

Coupon Rate < Market Yield

Price < Face Value

Example: A bond with 8% coupon priced to yield 10% will trade below its ₹1,000 face value — a discount bond.

Example 1

Indian Government Bond

10-Year G-Sec

YTM (Sep 2026)

=7.02%

India's benchmark 10Y G-Sec yield, reflecting current inflation and RBI policy expectations

Key Terms Illustrated

Nominal (Face) Value	₹100
Coupon Rate	7% p.a. (₹3.50 every 6 months)
Market Price	₹98 (trading at a discount)
Redemption	₹100 after 10 years
Yield to Maturity	7.02%

You lend ₹98 today → receive ₹3.50 every 6 months → get back ₹100 at year 10. Your YTM (7.02%) exceeds the coupon (7%) because of the capital gain from buying at a discount.

Key Terms Illustrated

Nominal (Face) Value	$1,000
Coupon Rate	4.5% p.a. ($22.50 every 6 months)
Market Price	$985 (trading at a discount)
Redemption	$1,000 after 10 years
Yield to Maturity	4.8%–5%

You lend $985 today → receive $22.50 every 6 months → get back $1,000 at year 10. Rising market yields push existing lower-coupon bonds to trade at a discount.

Example 2

US Government Bond

10-Year Treasury Note

YTM (Sep 2026)

=4.8%–5%

Near multi-year highs amid expectations of a Federal Reserve rate move

India vs. US: Side by Side

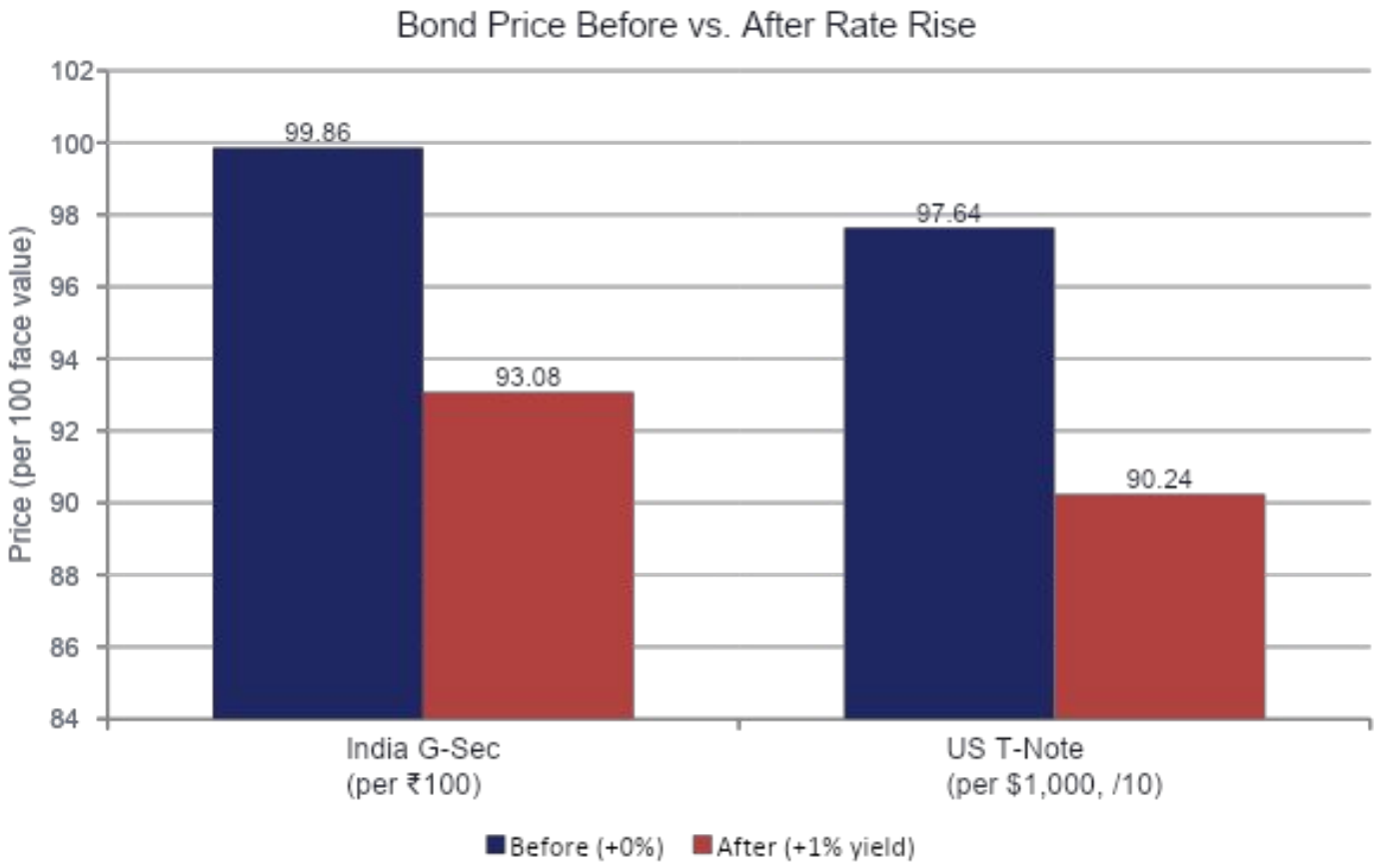
Same bond mechanics, different risk and rate environments

Term	Indian G-Sec	US T-Note
Face Value	₹100	$1,000
Coupon Rate	7%	4.5%
Redemption	₹100 @ Year 10	$1,000 @ Year 10
Yield to Maturity	7.02%	4.8–5%
Currency	Indian Rupee	US Dollar
Issuer Risk	Sovereign (India)	Sovereign (US)

Insight: India's higher coupon and YTM reflect greater inflation expectations and a country risk premium versus the US, which issues the world's reserve currency.

Interest Rate Risk: A +1% Yield Shock

Bond prices move inversely to yields — here's the impact on both 10-year bonds



India G-Sec

−6.79%

Price falls from ₹99.86 → ₹93.08

US T-Note

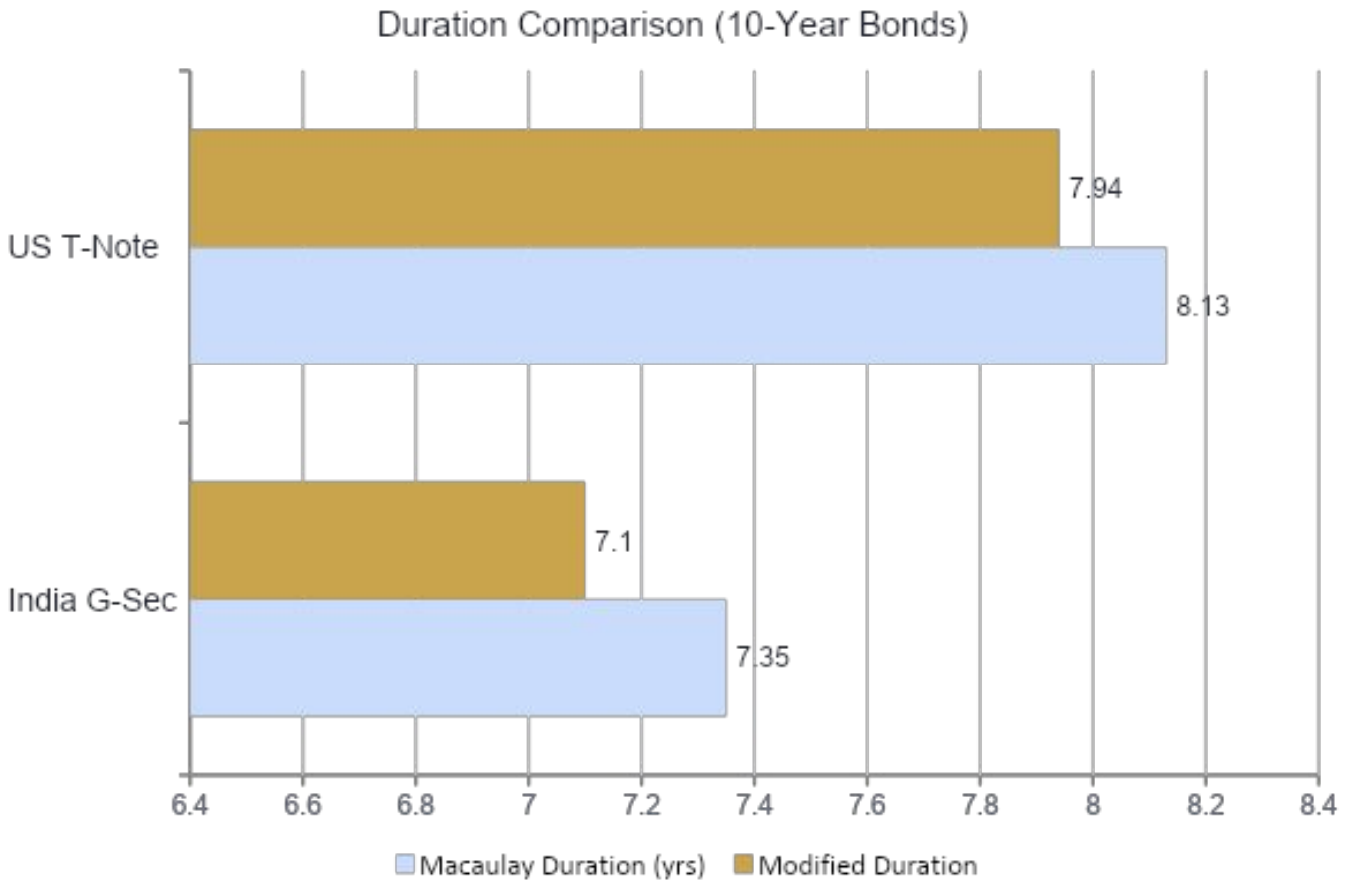
−7.58%

Price falls from $976 → $902

The US T-Note falls more — its lower coupon (4.5% vs 7%) means more of its value is "back-loaded" to maturity, making it more rate-sensitive.

Duration: Measuring Rate Sensitivity

Duration approximates the % price change for a 1% move in yield

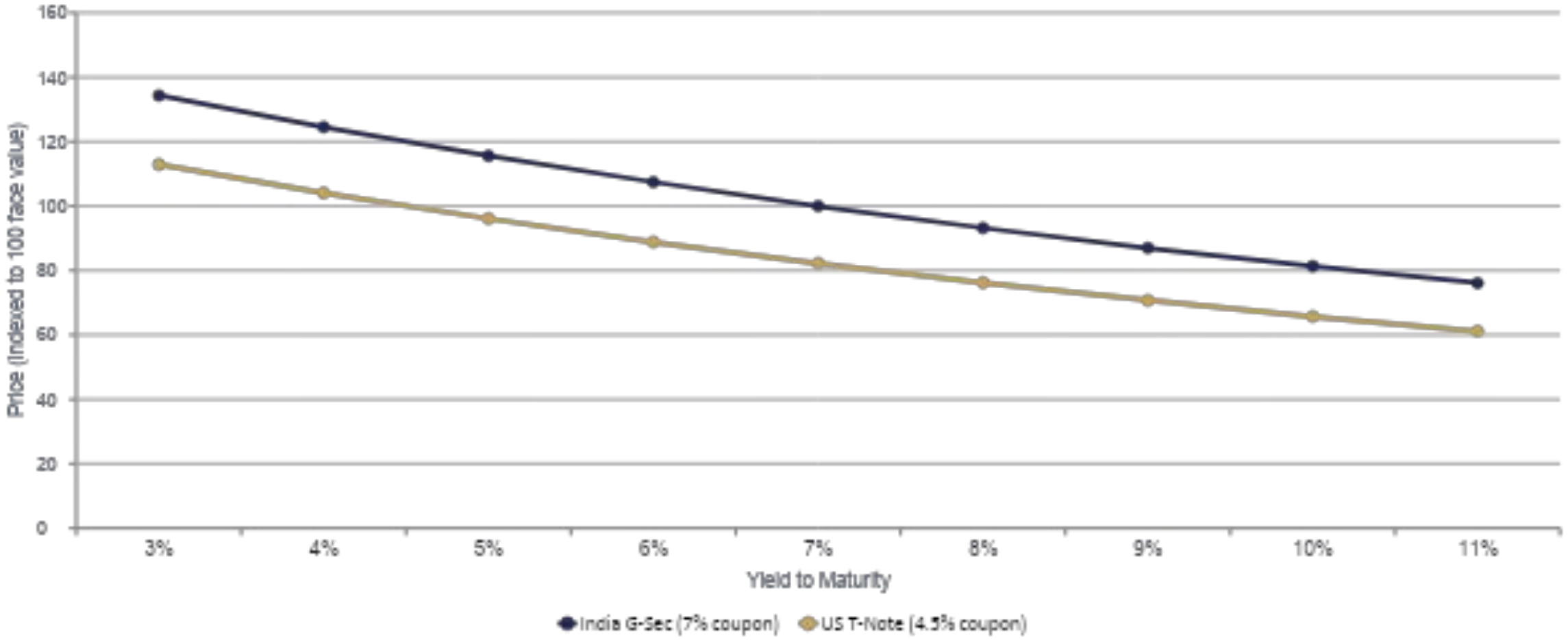


Why US Duration Is Higher

- Lower coupon (4.5% vs 7%) means less cash returned early via coupons
- More of total return is “back-loaded” to the final maturity payment
- Higher coupon → lower duration → less price sensitivity
- Longer maturity & zero-coupon bonds push duration even higher

The Price-Yield Curve: Why It Isn't a Straight Line

Bond prices fall as yields rise — but the relationship curves (convexity), so duration is only an approximation



Key Takeaways

Debt is a repayment obligation carrying interest, fixed tenure, and priority over equity.

A bond formalizes that debt — you lend to the issuer for coupons plus principal at maturity.

Coupon rate, face value, market price, redemption, and YTM define every bond's economics.

India's G-Sec (~7.02% YTM) yields more than the US T-Note (~4.8–5%) due to inflation & country risk premia.

Rates up ⟶ prices down. Lower-coupon, longer-duration bonds (like the US T-Note here) are more rate-sensitive.